

Contents

1	Appendix C: Detection Limits and Operating Points	1
1.1	Objective	1
1.2	Planned Methods (src)	1
1.3	Planned Script and Outputs	1
1.4	Figure	1
1.5	Equation References	1
1.6	Reproducibility	1
1.7	Cross-References	1

1 Appendix C: Detection Limits and Operating Points

1.1 Objective

Quantify minimum detectable power, SNR curves, and operating points for IR olfactory channels.

1.2 Planned Methods (src)

- `src/detection_limits.py`
 - `min_detectable_power(temperature_k, bandwidth_hz, snr_min_db)`
 - `snr_curve(signal_power_w, noise_temp_k, bandwidth_hz)`
 - `operating_point(capacity_bits_s, snr_db)`

1.3 Planned Script and Outputs

- Script: `scripts/generate_detection_limits.py`
- Data: `output/data/detection_limits.npz`
- Figure: `output/figures/detection_limits_operating_points.png`

1.4 Figure

1.5 Equation References

- Minimum power: see (??)
- Capacity: see (??)

1.6 Reproducibility

- Run: `python3 scripts/generate_detection_limits.py`
- Artifacts saved to `output/data/` and `output/figures/`.

1.7 Cross-References

- Methods: ??
- Symbols: ??

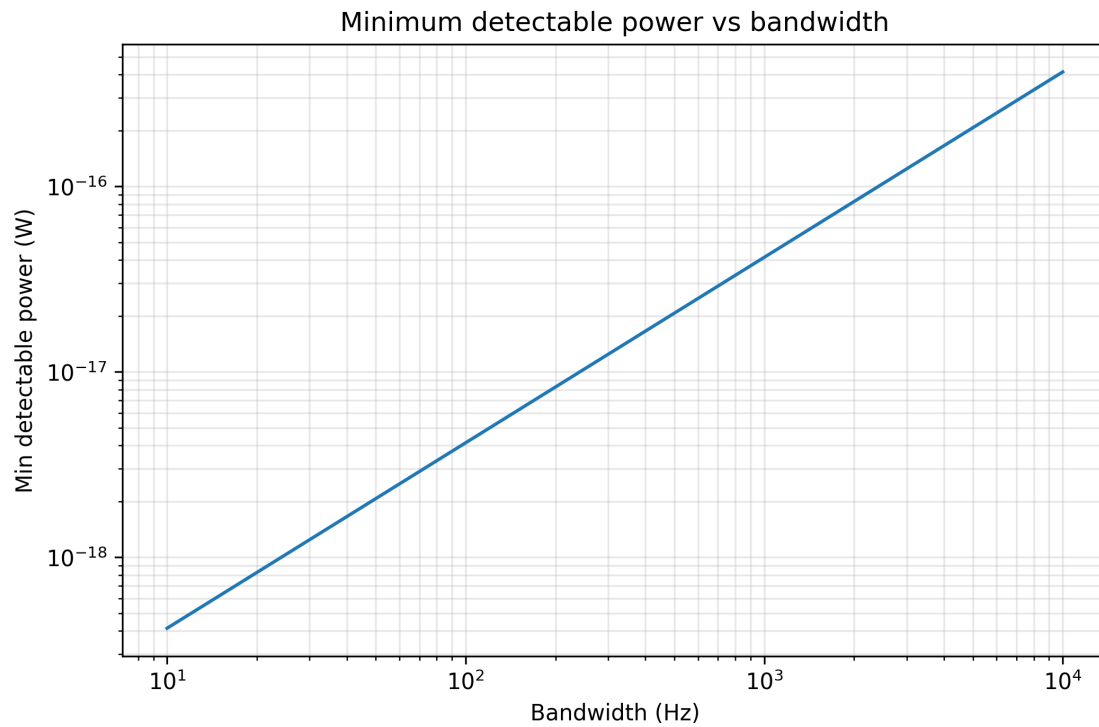


Figure 1: *Minimum detectable power contours and operating regions vs bandwidth and temperature; SNR and capacity overlays.*

- Math appendix: ??